Imaging & Tracking Objects Hidden Behind A Scattering Medium

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9th March 2026

Scattering of light leads to a degraded optical image, if enough light is scattered, only a speckle pattern can be measured. In this report it will be shown that the optical memory effect and computationally intensive numerical methods can be exploited to image an object that would otherwise be hidden. It will also be shown that the tracking of an object can be achieved by combining the autocorrelations of the speckle pattern at different times, which can be achieved using minimal computational effort.

1. Introduction

To see an image clearly, light is required to propagate from the object to your eye in a straight line. However, inhomogeneous media scatters this light, resulting in the image being distorted and/or blurred and for a strongly scattering media, it becomes impossible to filter out this unwanted signal.

Due to this, imaging through scattering media is an important problem, occurring in many areas of science. In biological imaging, for example, light passing through tissue is strongly scattered, this prevents conventional optical microscopes from being able to resolve structures within living organisms. A further example would be atmospheric imaging, where the atmosphere scatter incoming light, degrading observations.

Rather than attempting to prevent scattering, the approach used in this report aims to reconstruct an otherwise unusable image by taking advantage of the optical memory effect. This effect describes how the speckle pattern produced by coherent light passing through a scattering medium is strongly correlated with the speckle pattern produced from a different incident angle (within the memory effect range). This effect allows for the autocorrelation of the hidden object to be measured, which can be inverted using the Gerchberg-Saxton algorithm to obtain the shape of the object. This technique has a wide range of physical applications, the earlier examples being Stellar speckle interferometry [1], but it has also been applied to x-ray scattering [2] and optical microscopy [3]. As it will later be shown, this method is computationally intensive meaning it cannot be performed in real time for a moving object.

When tracking a moving object, it is not necessary to reconstruct a full image in order to understand its motion. Instead, by determining how the scene changes over time, it is possible to infer the movement of an object relative to a fixed background. This can be achieved by analysing the autocorrelation between measured intensity patterns at different times, which is less computationally intensive, allowing tracking to be performed in real time. However, as will later be explained, this method is only suitable for tracking a singular object.

The goal, that this paper will introduce, and further papers will hope to solve, is to be able to adapt the theory described above to allow tracking of two or more moving objects, using a singular method, which can be applied to any number of objects.

2. Imaging An Object

As light propagates through a scattering medium, it performs what can be thought of as a random walk (it can be shown that taking an average over all possible random walks yields the same result as the proper calculation [4]). As a result, the exiting light waves are out of phase and thus interfere with each other, producing a pattern of bright and dark spots known as a speckle pattern, a simulation of this is shown in Figure 1b. While the pattern appears to be random, they are in fact deterministic, depending on the precise configuration of the scattering medium and the illuminating wavefront. The interference pattern encodes information about the paths taken by the scattered waves as well as information about both the scattering medium and the light illuminating it. Analysing these patterns allows indirect information about the hidden object to be retrieved, which is the process that the following section describes.

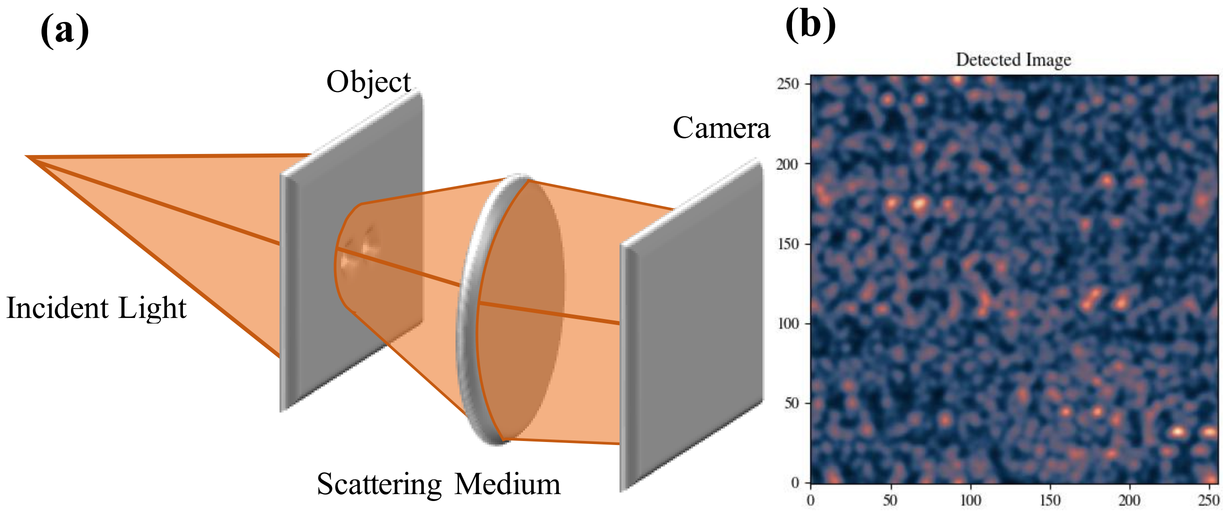

Figure 1 (**a**) An object (two holes) hidden behind a scattering medium emits light (because it is illuminated). The scattered light forms a speckle pattern (**b**), which is detected by a camera (simulation). [5]

The Optical Memory Effect

The first step in extracting the information encoded within the speckle pattern is to look at the correlation between the intensity due to two different incident waves. For two incident plane waves with wavevectors $\boldsymbol{k}_a$ and $\boldsymbol{k}_{a'}$ the correlation between the intensity in the emerging directions $\boldsymbol{k}_b$ and $\boldsymbol{k}_{b'}$ is given by [4]

$$C^{I}_{a,b,a',b'} = \frac{\langle \delta I_{a,b}\, \delta I_{a',b'} \rangle}{\langle I_{a,b} \rangle \langle I_{a',b'} \rangle}, \tag{1}$$

where $\delta I = I - \langle I \rangle$ and $\langle\ .\rangle$ represents an average. From there, it can be shown that the leading term of this correlation has the form [4]

$$C^{(1)}_{a,b,a',b'} = \delta(\Delta k_a - \Delta k_b)\left(\frac{|\Delta k_a| L}{\sinh(|\Delta k_a| L)}\right)^2 \tag{2}$$

where $\Delta k_a = \boldsymbol{k}_a - \boldsymbol{k}_{a'}$, δ is the Dirac delta and L is the thickness of the scattering medium. Therefore, if the angle of incidence is changed by a small amount $|\Delta k_a|$ such that $\left(\frac{|\Delta k_a| L}{\sinh(|\Delta k_a| L)}\right)^2 \sim 1$ the speckle pattern for the wave a will be highly correlated with the pattern for the wave a' (i.e. the patterns will look the same). This is significant over an angular range of the order λ/L. Within this range, a shift in the incident beam causes the speckle pattern to shift by the same amount, $|\Delta k_b| \simeq |\Delta k_a|$. This demonstrates the remarkable concept that, beyond the multiple scattering process, the emerging beam retains some information about the incident beam.

Speckle Interferometry

For an object emitting (or reflecting) light behind a scattering medium, each point $\boldsymbol{x}_0$ will radiate an amount $O(\boldsymbol{x}_0)$ of light towards the scattering screen. For narrow-band detection, the light coming from each point will create a speckle pattern $S(\boldsymbol{x}_0, \boldsymbol{x}_d)$ at position $\boldsymbol{x}_d$ on the detector. For low spatial coherence, these speckle patterns will not interfere, they will sum, giving a total measured intensity on the detector [5]

$$I(\boldsymbol{x}_d) = \int O(\boldsymbol{x}_d) S(\boldsymbol{x}_0, \boldsymbol{x}_d) d^2\boldsymbol{x}_0 \, . \tag{3}$$

The goal is to retrieve the object, O and to do this the mathematical technique autocorrelation is used, which is an operation that measures how similar a signal is to a shifted version of itself. The autocorrelation, denoted by $\star$, of the measured intensity is taken. From there it can be shown that [5]

$$\langle I \star I \rangle (\Delta \boldsymbol{x}_d) \propto [O \star O] \otimes C^{(1)} \tag{4}$$

where $\otimes$ is the convolution product. This relation holds within the limit $\frac{|\Delta x_0|}{d} \sim \frac{\lambda}{L}$, where d is the distance between the object and the scattering medium.

Eqn. (4) relates the autocorrelation of the image to the autocorrelation of the object; however, autocorrelation is a lossy operation meaning that it cannot be inverted to find O. For a simple object with a known autocorrelation, Eqn. (4) is all that is needed. For example, if the object is a binary system (often found in astronomy as binary star systems) with separation d then the autocorrelation of this would be three small dots, where the separation between the central and side dots is d. The central dot is brighter, representing the perfect overlap between the image and the shifted version, for a shift of zero. The side dots represent when a shift of d to the left or right causes one of the shifted dots overlaps with an original, see a simulation of this shown in Figure 2.

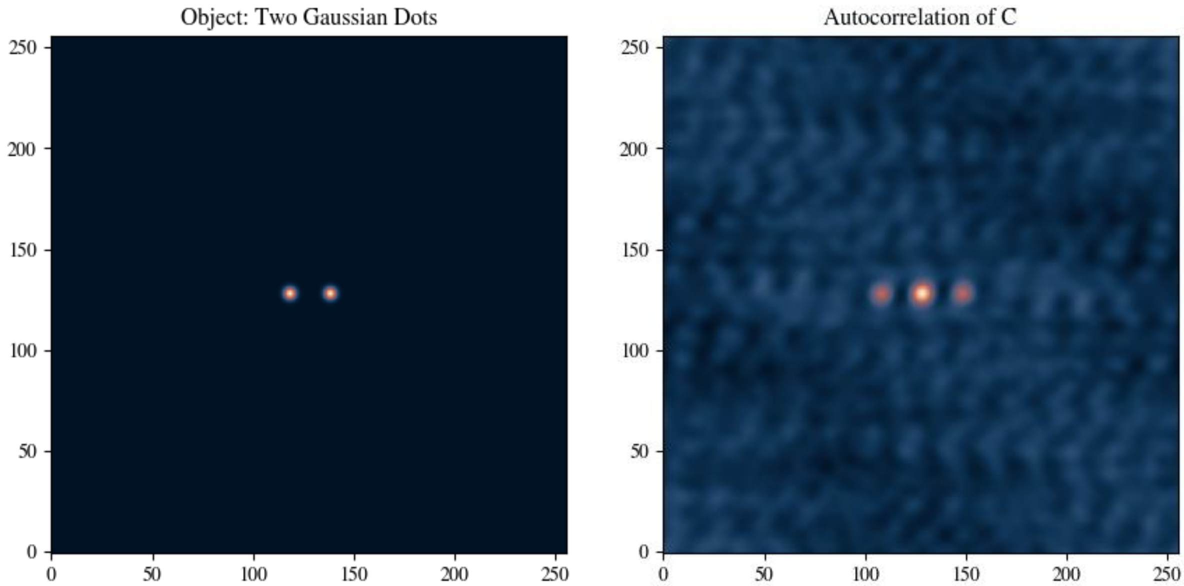

Figure 2 Left: a binary system, two Gaussian dots. Right: autocorrelation of two Gaussian dots, with the central dot being brighter. (simulation)

Phase Retrieval & The Gerchberg–Saxton Algorithm

To retrieve the information associated with O, the object is assumed to be real, positive and to occupy a finite region. Due to the object being two-dimensional (or higher), the solution found by the following technique is unique, which is what allows this algorithm to work effectively. The first step is to use the Wiener-Khinchin theorem to relate Fourier transform of an autocorrelation to the modulus squared of a Fourier transform, [4]

$$\mathcal{F}[O \star O] = |\mathcal{F}[O]|^2 \quad (5)$$

where $\mathcal{F}$ represents the Fourier transform. This provides a way to extract the modulus of the Fourier transform of the object; however, due to the modulus, the phase of the Fourier transform is still unknown and will have to be approximated using the Gerchberg–Saxton algorithm.

The Gerchberg–Saxton algorithm is based on the idea that all phases, except the correct one, violate the initial assumptions stated above. The algorithm starts with an initial guess for the object, $g_k(r)$, the Fourier transform is taken, $\tilde{g}_k(k)$, then the incorrect amplitude is substituted with the correct one retrieved by Eqn. 5, this is denoted by $\tilde{g}_k{}'(k)$.

From there, iteration is then used to improve the guess, where every point that does not meet the constraints is set to zero, [4]

$$g_{k+1}(r) = \begin{cases} g_k'(r) & \text{if constrains satisfied} \\ 0 & \text{else} \end{cases}. \tag{6}$$

This algorithm is then repeated until the function produced is satisfactory, describing the initial object to a given accuracy. However, the algorithm has a tendency to converge slowly (this can be made slower by a poor choice of initial function) and get stuck for long periods of time. While there are algorithms that improve this, such as the hybrid input-output algorithm, they are outside the scope of this report, see the following paper for more information [4].

One thing to note is that autocorrelations do not contain any information about absolute position, just relative position between points. Furthermore, the autocorrelation of a real function is centrosymmetric, so the algorithm cannot distinguish between the image of the object and the same image flipped. This becomes more important when looking at tracking an object.

3. Tracking An Object

If all the points in the scene moves rigidly (i.e. the scene is translated, not deformed), then the movement can be reconstructed by taking the autocorrelation of the light measured at time t_1 with the light measured at time t_0. It is worth noting that the times do not have to be sequential and can instead be multiple frames apart. Due to the optical memory effect, this results same autocorrelation $O \star O$, just shifted by the amount the scene moved within the time interval.

Often this not the case such as for a single object moving relative to a fixed background. Taking the autocorrelation produces two terms; first term describes how much the given object has moved, the second term contains all cross-correlations between all the objects in the scene, including information about the background objects that never moved. This extra information dominates the output, making the object very hard to track.

To mitigate this effect, $I(t_1) \star I(t_0) - I(t_1) \star I(t_1)$ is calculated instead [5]. This removes most of the information associated with the static objects by subtracting the autocorrelation of the current scene. The only information left is that on their relative distance from the moving object. This calculation produces a result in the reference frame of the object; the object is always in the centre of the image, with the static background moving around it, following the trajectory of the initial movement. In practice, this is one of the reasons why tracking two or more objects is unachievable using this method as each object tracked will result in that object having its own moving background showing the trajectory. This complicates the scene making it very hard for a human, and especially a computer, to separate the set of moving backgrounds. There will also be a 'ghosting' effect with each dot will being a bright-dark pair, with the leading dot showing the direction of motion. For a demonstration of this and to see a tracking simulation, see GitHub repository - supplementary videos [6].

4. Discussion

Imaging in scattering media does not have one unified solution, but rather different techniques that work under different conditions. The technique used in this report is based on the optical memory effect and how the autocorrelation of the object is numerically inverted using the Gerchberg–Saxton algorithm. While this technique works for strongly scattering medium (providing the signal output is strong enough) [4], the phase retrieval is computationally intensive, though it could be sped up significantly by processing the various Fourier transforms in such a way to take advantage of modern computer hardware. The algorithm is also affected by the number of iterations performed and the initial guess used. The use of artificial intelligence, more specifically machine learning, may prove an approach that develops initial guesses that are more likely to lead to a highly accrual result while using less iterations. The other limitation is due to the angular range of the optical memory effect. The speckle correlation due to the optical memory effect decreases rapidly if $|\Delta k_a|L \gg 1$, meaning the objects being imaged will need to be very small or very far away from the scattering layer.

Tracking the movement of an object, by comparison, has a rather simple solution; an appropriate combination of the cross-correlations of the speckle pattern at different times. The output of this is both human-interpretable and possible to run in real time, with minimal computation, and without any need for any previous knowledge about the object (such as that used to inform the guess used in the Gerchberg–Saxton algorithm). This method can track lateral translations, as well as rotations, changes of size, deformations, and any other type of movement. However, this is all limited by the memory range and the amount of signal available. The range of the memory effect decreases exponentially with the thickness of the scattering medium L as shown in Eqn. 2. This means that for scattering mediums on the order of a wavelength (or thicker), the object must move less than the memory range between two successive frames. Furthermore, if the object did not move by a great enough distance, only a fraction of the light is able to pass through the scattering medium and reach the camera. Therefore, sensitive and fast detectors with a good dynamic range will be required for real-world applications, for more information see [7].

5. Applications

The default application for any imaging technique is almost always astronomy, and this is indeed where this field was first developed. Known as Stellar speckle interferometry, it was used to increase the resolution of telescopes beyond the limitation dictated by seeing, a topic explored in the following paper [1]. A similar application is seen in the field of X-ray imaging. Previously, X-ray imaging could only be used for crystalline structures, however, analysis of the speckle patterns produced by imaging non-crystalline specimens now means that X-ray imaging can be used for both crystalline and non-crystalline structures, filling the gap between visible light microscopy and electron microscopy, this is covered in greater detail in [2].

Another application is non-invasive imaging; when viewed from a medical standpoint this provides the ability to image, and potentially track, an object inside the human body without having to remove the scattering medium (e.g. tissue). An example of this being used in practice comes from the paper "Non-invasive imaging through opaque scattering layers" [3] where a sample from the stem of Lily of the valley (*Convallaria majalis)* was illuminated by diffused light, the autofluorescent sample emitted light that then passed through a scattering material (placed 6 mm away) before being recorded. After an average was taken across five scans, all the high-intensity features in the original object were recovered, Figure. 3.

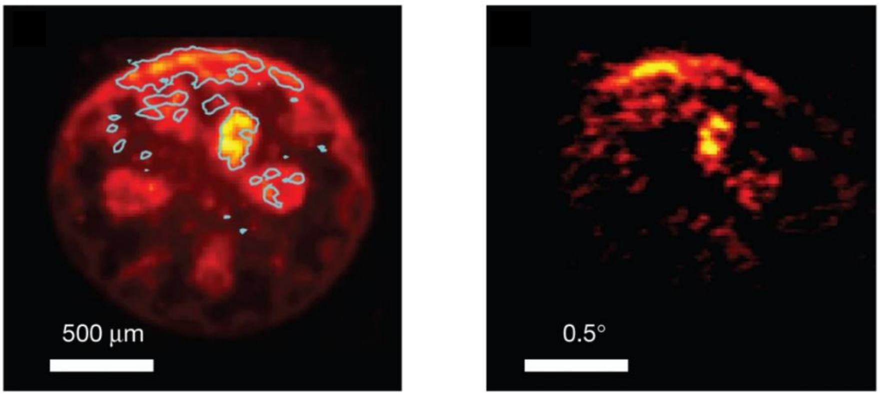

Figure 3 Left: Fluorescence image of the structure taken after removing the back diffuser and averaging over many (~200) speckle illuminations. Right: The reconstructed object.

6. Looking Ahead

The goal of the project associated with this report is twofold; firstly, to develop a method to be able to track two moving objects and secondly, to have this method work for more than two moving objects. To achieve this, the first step is to replicate previous work on the topic, using both computer simulations (shown in Figure. 1 and Figure. 2 as well as the supplementary videos [8]) and by recreating experimental data, such as that described in [4]. This will help to identify any issues with the experimental setup and data analysis, as the experiment has known outcomes, and also provide a framework which can be adapted when looking at two or more objects.

An example experimental setup to image an object would consist of a coherent light source, a 1 mW CW polarised HeNe laser; a beam expander, made from two lenses with focal length –25mm and 300mm; a spinning diffuser, to reduce spatial coherence of the beam; an object, aluminium foil with a shape made of pinprick holes; scattering medium, 220 grit ground glass optical defuse and a camera, which needs to have at least a 12-bit dynamic range [4].

From there, the task will be to develop the theory on how to track multiple objects. The starting point will be to construct a mathematical method to analyse the speckle produced by multiple objects, this will then be translated into a piece of code to achieve this computationally. From there, the theory will be tested by both simulations and by developing and performing a succession of experiments with the overall goal to be able to track any number of moving objects, providing there is a static background, using a single methodology.

7. References

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